

Assume X is CRV and Y is DRV and X and Y are independent

Then CDF of $Z = X + Y$ is

$$F_Z(z) = P(X + Y \leq z)$$

$$= \sum_{y \in S_Y} P(X + Y \leq z | Y = y) f_Y(y)$$

$$= \sum_{y \in S_Y} P(X \leq z - y) f_Y(y)$$

$$= \sum_{y \in S_Y} \int_{-\infty}^{z-y} f_X(x) f_Y(y) dx$$

$$= \sum_{y \in S_Y} \int_{-\infty}^z f_X(x' - y) f_Y(y) dx', \text{ taking } x' = x + y$$

$$= \int_{-\infty}^z \left(\sum_{y \in S_Y} f_X(x' - y) f_Y(y) \right) dx'$$

$$\text{Thus } f_Z(z) = \sum_{y \in S_Y} f_X(x' - y) f_Y(y) \quad \forall z \in \mathbb{R}.$$